

Ankit Kumar

Machine Learning Engineer | Agentic AI, RAG & Production ML Systems

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Seattle, WA | F-1 STEM OPT through May 2028

Machine Learning Engineer with 4+ years across industry and venture work, building ML and GenAI systems spanning hybrid RAG, retrieval and reranking, physics-constrained modeling, and embedded inference. Owns systems from ambiguous requirements through evaluation, deployment, and cross-functional validation.

EXPERIENCE

Milwaukee Electric Tool Corporation

July 2025 - July 2026

Machine Learning Engineer - Front End Innovation (FEI)

- Led development of a physics-informed PyTorch thermal warning system combining a three-body ODE with electrical heat generation; achieved a worst-case RMSE of 2.2°C across 30 held-out profiles and warnings up to 60 seconds before hard shutoffs.
- Cut experiment iteration from two hours to 40 minutes by building Azure ETL and training pipelines that synchronized lab and field data, automated preprocessing and model execution, and versioned datasets, parameters, metrics, and artifacts in MLflow.
- Reduced validation time from 30 days to three by validating laboratory load-brake cycles against field-use heat-generation profiles, establishing a repeatable substitute for hands-on cutting tests.
- Identified a sensor-induced 4-5°C model error through ablation and counterfactual testing from -20°C to 40°C, supporting its removal and \$100K in projected hardware savings.
- Partnered with the principal firmware engineer to port the PyTorch model to embedded C and built a time-causal Python reference twin with Streamlit telemetry replay; caught two release-blocking defects in OCV/DCIR handling and model initialization before field validation, avoiding an estimated 30+ days of rework.

Mitra | Mason Innovation Exchange, GMU

June 2024 - May 2025

Founder & Lead ML Engineer - Selected University Venture

- Led end-to-end development of an AI-powered university decision-support platform that earned 78% positive ratings in professor-led classroom evaluations involving 200+ students.
- Built an event-driven Scrapy, AWS, and PySpark pipeline processing 8K+ webpages per crawl; filtered approximately 70% of low-value pages before embedding and maintained a source-traceable corpus of 50K+ chunks through approximately 30-minute Glue ETL runs.
- Improved top-5 retrieval hit rate from 42% to 87% on a fixed labeled GMU query set by combining MiniLM embeddings, FAISS HNSW, lexical search, weighted reciprocal rank fusion, and cross-encoder reranking.
- Designed a bounded LangGraph workflow for ambiguous and multi-intent queries, limiting each request to five agent actions and two LLM calls; retried retrieval, requested clarification, or abstained when evidence was insufficient, while monitoring SageMaker-hosted retrieval and reranking services through LangSmith and CloudWatch.
- Raised precision@20 from 25% to 45% across two offline evaluation cycles by validating user feedback with DistilBERT, retraining the pairwise cross-encoder over a larger candidate pool, and versioning checkpoints for rollback.

Accenture

February 2021 - July 2023

Packaged App Development Analyst

- Built PySpark and SQL pipelines on Amazon S3 that standardized point-in-time fundamentals for approximately 9,500 public companies into partitioned Parquet, reducing company and competitor-analysis preparation from 5 business days to 1.5 days.
- Developed pooled LSTM/XGBoost and per-company SARIMA models for two-year KPI forecasts, improving stable-regime directional accuracy from a 64% persistence baseline to 78% and transition-period accuracy from 52% to 61%.
- Fine-tuned FinBERT on approximately 75K earnings calls, news articles, and analyst reports using subsequent price movement as supervision; achieved 0.76 macro-F1 on a chronological holdout and improved transition-period directional accuracy from 61% to 66%.

PROJECTS

PyLinguist: Cross-Lingual Python Translation

November 2024 – Present

Research project | Submitted to ACL BEA

- Built a two-stage Python translation system across seven languages using token-aware translation, a curated dictionary of 234 Python terms, and Google Translate followed by GPT-4o or Claude 3.5 Haiku refinement while preserving code syntax and structure.
- Built a back-translation evaluation framework spanning BLEU, syntax, structural preservation, semantic similarity, and bilingual review; across 70 held-out programs, GPT-4o achieved a 92.5% composite score versus a 61.5% baseline and maintained 100% syntax validity.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C, PySpark, ETL, Parquet

GenAI & Retrieval: RAG, LangGraph, LangChain, Pydantic, Prompt Engineering, Evidence-Gated Generation, SentenceTransformers, FAISS HNSW, Hybrid Search, Weighted RRF, Cross-Encoder Reranking

ML & Evaluation: PyTorch, XGBoost, LSTM, SARIMA, FinBERT, DistilBERT, Time-Series Forecasting, LLM/Agent Evaluation, Model Validation, BLEU

Cloud & MLOps: AWS (SageMaker, Lambda, S3, SQS, Glue, Athena, CloudWatch), Azure, MLflow, LangSmith, Streamlit

EDUCATION

George Mason University

August 2023 - May 2025

M.S. in Computer Science, Machine Learning Concentration | GPA: 3.90/4.00

Courses: Mining Massive Datasets with MapReduce, Advanced NLP, Machine Learning, Data Mining, Artificial Intelligence

Achievements: Distinguished Academic Achievement Award by College of Engineering and Computing

CERTIFICATIONS

- AWS Certified Machine Learning Engineer - Associate (MLA-C01)
- AWS Certified Developer - Associate (DVA-C02)